
MA 565: Numerical Methods in Quantitative Finance

Course Code	MA 565
Course Name	Numerical Methods in Quantitative Finance
L-T-P-C	3-0-0-3
Prerequisites	IC 110: Engineering Mathematics, IC 111: Linear Algebra
Intended for	UG/PG
Distribution	Elective
Approval	8th Senate (Oct 2015)

Course Contents

- **THE CONTINUOUS THEORY OF PARTIAL DIFFERENTIAL EQUATIONS** [8 Lectures]
An Introduction to Ordinary Differential Equations, An Introduction to Partial Differential Equations, Second-Order Parabolic Differential Equations, An Introduction to the Heat Equation in One Dimension, An Introduction to the Method of Characteristics.
- **FINITE DIFFERENCE METHODS: THE FUNDAMENTALS** [8 Lectures]
An Introduction to the Finite Difference Method, An Introduction to the Method of Lines, General Theory of the Finite Difference Method, Finite Difference Schemes for First Order Partial Differential Equations, FDM for the One Dimensional Convection Diffusion Equation, Exponentially Fitted Finite Difference Schemes.
- **APPLYING FDM TO ONE FACTOR INSTRUMENT PRICING** [8 Lectures]
Exact Solutions and Explicit Finite Difference Method for One Factor Models, Exponentially Fitted Finite Difference Schemes for Barrier Options, Advanced Issues in Barrier and Lookback Option Modelling.
- **FDM FOR MULTIDIMENSIONAL PROBLEMS** [8 Lectures]
Finite Difference Schemes for Multidimensional Problems, Operator Splitting Methods: Fractional Steps, ADI Methods.
- **APPLYING FDM TO MULTI FACTOR INSTRUMENT PRICING** [10 Lectures]
Options with Stochastic Volatility: The Heston Model, Finite Difference Methods for Asian Options and Other 'Mixed' Problems.

Text Books

1. Daniel J. Duffy, **Finite Difference Methods in Financial Engineering: A Partial Differential Equation Approach**, John Wiley & Sons Ltd., 2006.
2. D. Tavella and C. Randall, **Pricing Financial Instruments: The Finite Difference Method**, Wiley.

References

1. Paolo Brandimarte, **Numerical Methods in Finance and Economics: A MATLAB Based Introduction**, 2nd Edition, John Wiley & Sons, 2006.
2. John A. D. Appleby, David C. Edelman, John J. H. Miller, **Numerical Methods for Finance**, Taylor & Francis, 2008.
3. Michele Breton, Hatem Ben Ameer, **Numerical Methods in Finance**, Springer, 2005.